Embedded Controller

USART: Serial Communication Protocol

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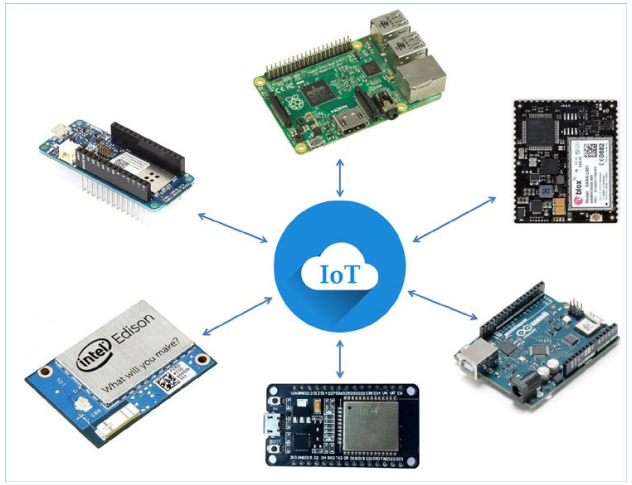
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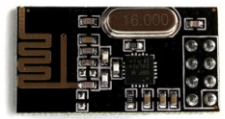
Introduction

- **Wired Communication (serial)**
  - UART
  - Ethernet 10/100/1000
  - USB

- **RF Communication**
  - 20kHz~300GHz
  - Bluetooth (2.4GHz)
  - WiFi (2.4, 5, 6 GHz)
  - Zigbee (2.4GHz)
  - Radio Control (900MHz, 2.4GHz)
  - LoRA (<1GHz)
  - NFC
  - 3G/LTE/5G

- **Examples of IoT modules**
  - MKR WiFi1010, MKR WAN 1300
  - ESP32/ESP8266 (WiFi)
  - nRF24L01 (2.4 RF)
  - HC-12 433MHz
  - HC-06 (BLE)









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Introduction

- What is communication protocol?
  - A system of description of digital data format and rule for communication

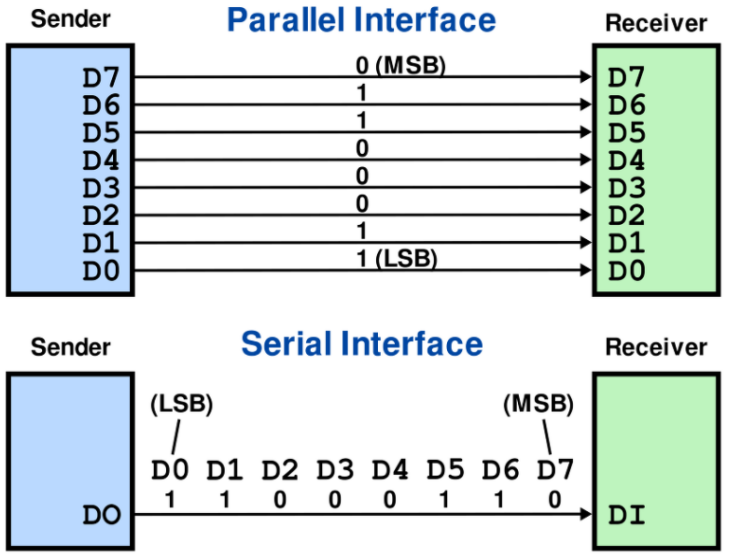
- We will cover protocols of **UART, SPI, I2C**
  - **UART:** Universal Asynchronous Receiver/Transmitter
  - **SPI:** Serial Peripheral Interface (synchronous)
  - **I^2C**: Inter-Integrated Circuit

Asynchronous: UART

Introduction

- **How to transmit/receive 8-bit data?**
  - Serial I/O : 1 bit at a time
  - Parallel I/O: multiple bits at once

- **Serial I/O**
  - Fewer wires but fast enough for many application
  - Common Serial Standards
    - **UART(or SCI):** Universal Asynchronous Receiver/Transmitter
      - SCI: Serial Communication Interface
      → RS-232, RS-422, RS-485
    - **SPI:** Serial Peripheral Interface
    - I^2C: Inter-Integrated Circuit
    - Ethernet
    - USB and more



* RS: Recommended Standard

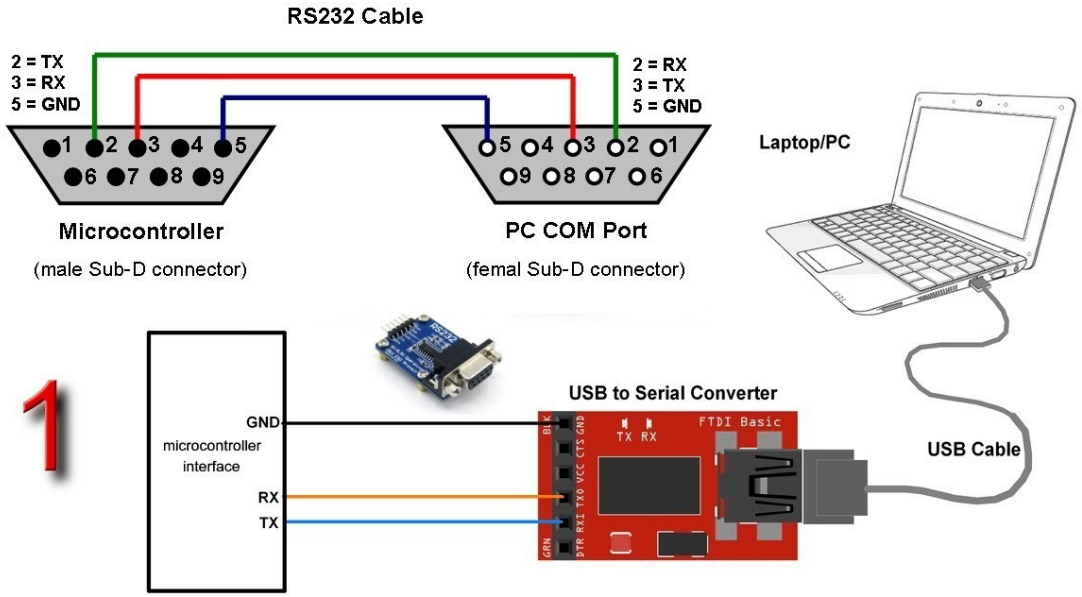
UART Introduction

- **UART(Universal Asynchronous Receiver/Transmitter)**
  - UART itself is not a communication protocol like SPI and I2C
  - **It is a physical circuit** (convert 8-bit parallel to serial data)
  - Initially for Teletypewriter-Modem
  - Later, RS-232 compatible became standards for serial communications for PC
    - It is replaced by USB in modern PC

  - Standards
    - RS-232 (1969)
    - RS-422 (apple), RS-485

      * RS (Recommended Standard)

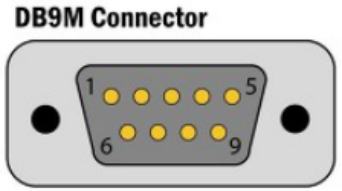




UART Introduction

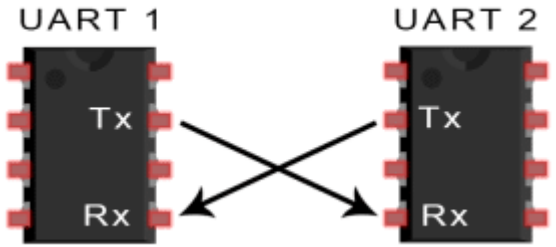
- **UART(Universal Asynchronous Receiver/Transmitter)**
  - Asynchronous transmission:
    - Serial I/O that communicates without common 'CLOCK'
      > Transmitter, Receiver uses its own clocks
    - But, agree on transmission rate (Baud Rate)

  - Characteristics
    - Simple to configure, can use two wires, no connected CLOCK
    - Can use a parity bit to allow for error checking
    - No Multiple Master/Slave
    - Relatively slow
    - Weak to noise/short range



RS232 Pin Out

Pin #	Signal
1	DCD
2	RX
3	TX
4	DTR
5	GND
6	DSR
7	RTS
8	CTS
9	RI



UART Introduction

- **UART**
  - Sending a character (data packet) at the given Baud Rate
  - **Baud rate:** signaling rate of symbols/second
    - Baud rate: 9600, 14400, 19200, 38400, 57600, 115200
    - A symbol is a group of bits (e.g. 10 bits: Start+8bit+Stop)
    - **Data rate:** bits/seconds (bps)
  - **Start** and **Stop** bits
  - **Parity bit** can be added optionally

1 symbol(10bits) example:

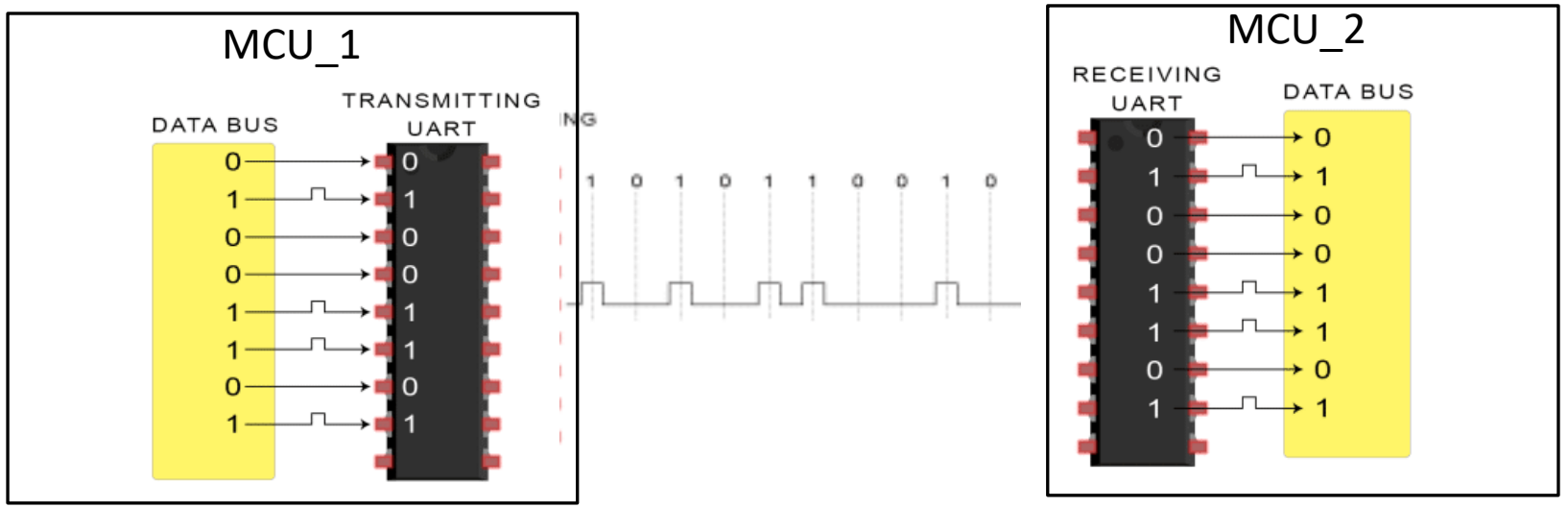
8-bit data(character)+ Start bit+ Stop bit+ no_parity

9,600 baud rate → 9600 (symbols/s) * (10bits/symbol)= 96,000 bps



UART Introduction

- **UART**
  - Receiver/Transmitter must have the same configuration (protocol)
  - Need to check the voltage level for HIGH (5V, 3.3V etc) and LOW
    - STM32F4 gives H=3.3V
  - Add Start(0), Parity, Stop(1)
  - Data Parallel → UART(TX) → Serial → UART(RX) →Parallel Data

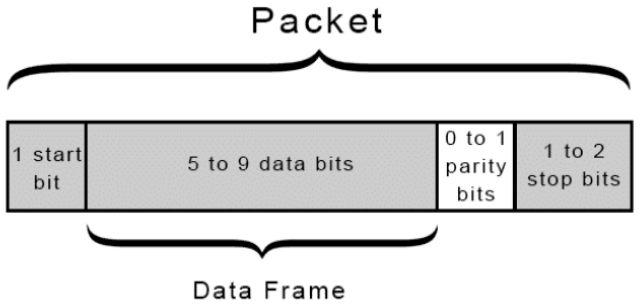


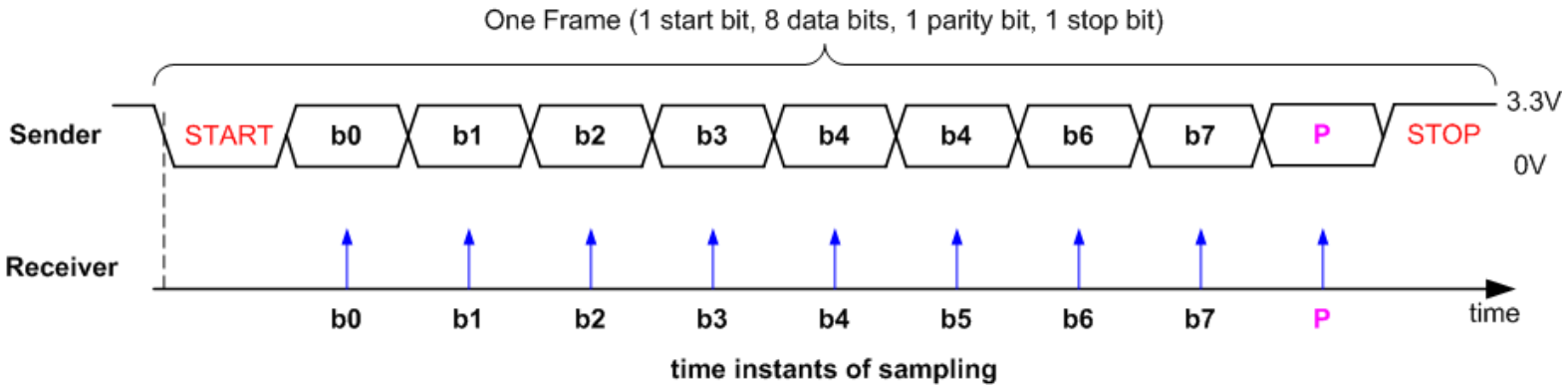
Parallel from data bus | Serial Transmission | Parallel to data bus

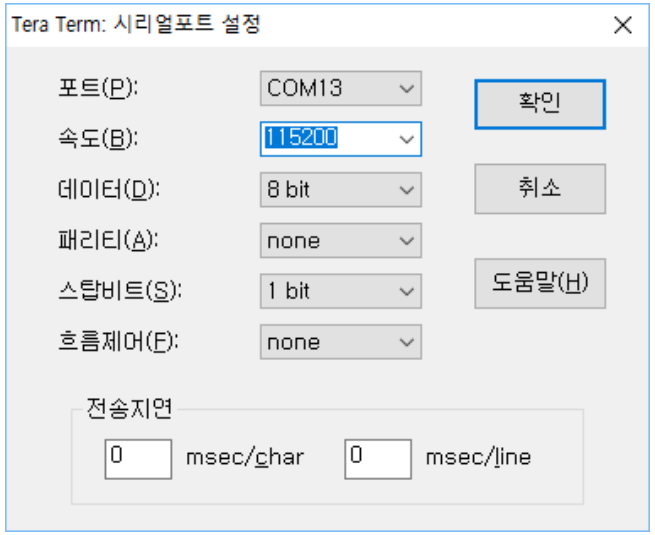
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UART Introduction

- **UART**
  - Data Packet: Idle(1bit) + Start(1bit) + Data (7~8bits) + Parity(option) + Stop(1, 1.5, 2 bits)
  - Idle: H when not in use.
  - Start: H->L starts transmission
  - Data: Least significant bit sent first
  - Parity: None/Even/Odd parity
    - > number of H bits to make it Even or Odd
    - > e.g. Data= 011 odd parity: parity bit =1
  - Stop bits: L → H for 1-bit or 2-bits



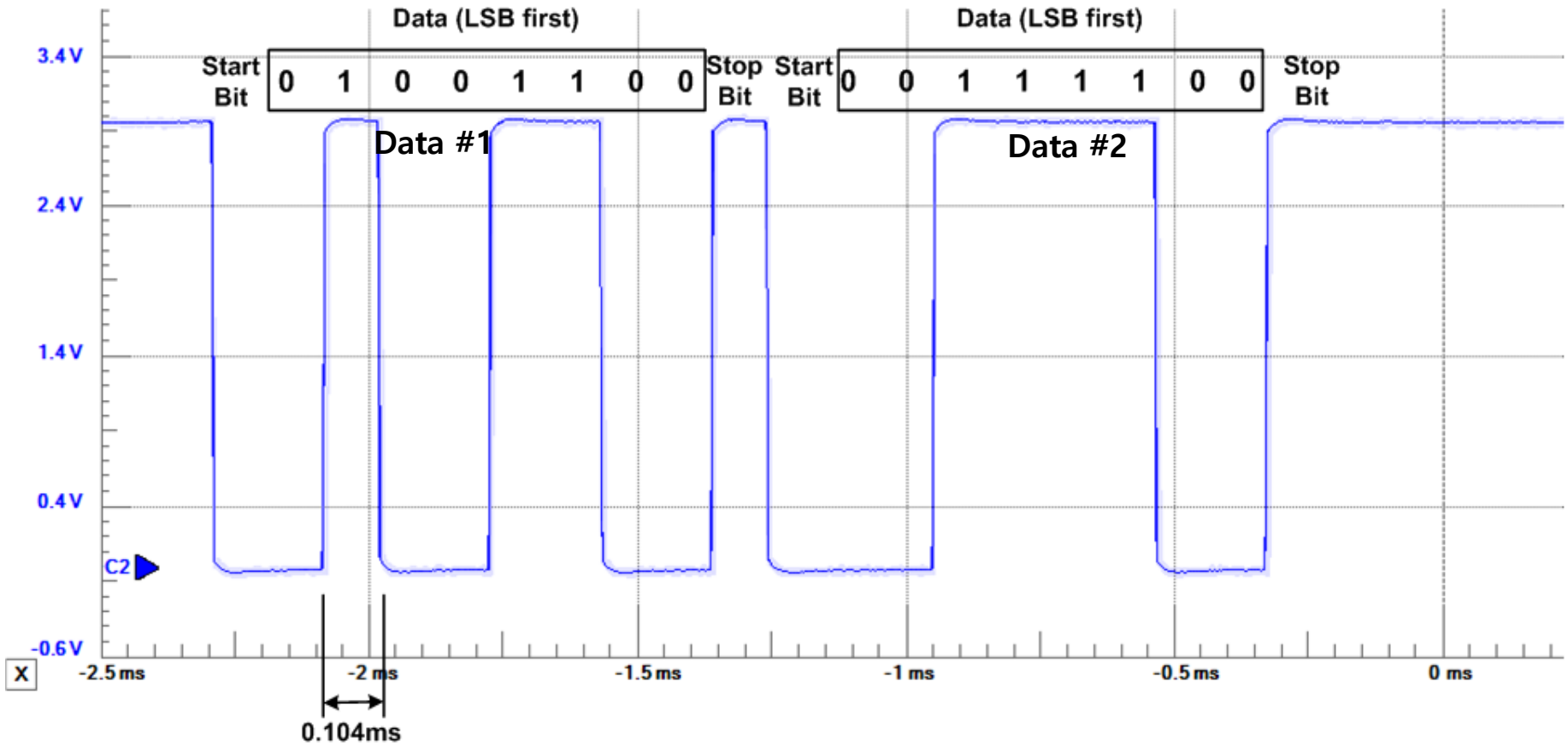




UART Introduction

- **Example:**

  Data #1: Transmitting 0x32 = 0011 0010
  Data #2: Transmitting 0x3C = 0011 1100



1 start bit, 1 stop bit, 8 data bits, no parity, baud rate = 9600

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UART: RS-232

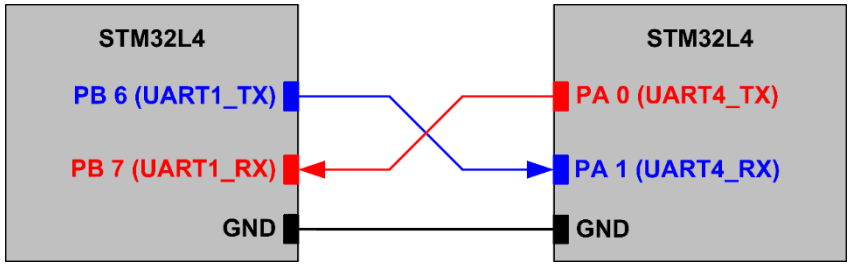
- **UART standards**
  - UART has several standards

Characteristics of RS232, RS422, RS423 and RS485

	RS232	RS423	RS422	RS485
Differential	no	no	yes	yes
Max number of drivers Max number of receivers	1 1	1 10	1 10	32 32
Modes of operation	half duplex full duplex	half duplex	half duplex	half duplex
Network topology	point-to-point	multidrop	multidrop	multipoint
Max distance (acc. standard)	15 m	1200 m	1200 m	1200 m
Max speed at 12 m Max speed at 1200 m	20 kbs (1 kbs)	100 kbs 1 kbs	10 Mbs 100 kbs	35 Mbs 100 kbs
Max slew rate	30 V/µs	adjustable	n/a	n/a
Receiver input resistance	3..7 kΩ	≧ 4 kΩ	≧ 4 kΩ	≧ 12 kΩ
Driver load impedance	3..7 kΩ	≧ 450 Ω	100 Ω	54 Ω
Receiver input sensitivity	±3 V	±200 mV	±200 mV	±200 mV
Receiver input range	±15 V	±12 V	±10 V	-7..12 V
Max driver output voltage	±25 V	±6 V	±6 V	-7..12 V
Min driver output voltage (with load)	±5 V	±3.6 V	±2.0 V	±1.5 V

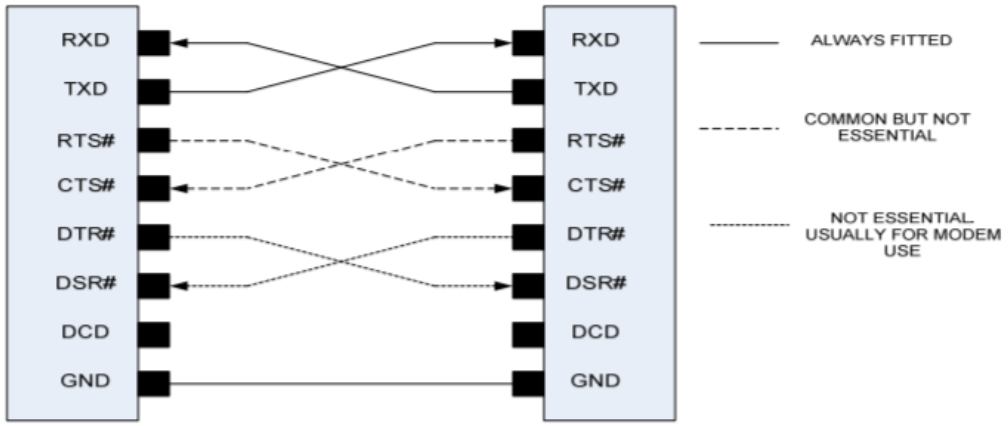
UART: RS-232

- **UART: RS-232**
  - MCU typically use simplified 3-pins
  - (GND, TX, RX), and possibly (RTS and CTS)



Complete RS-232 pins

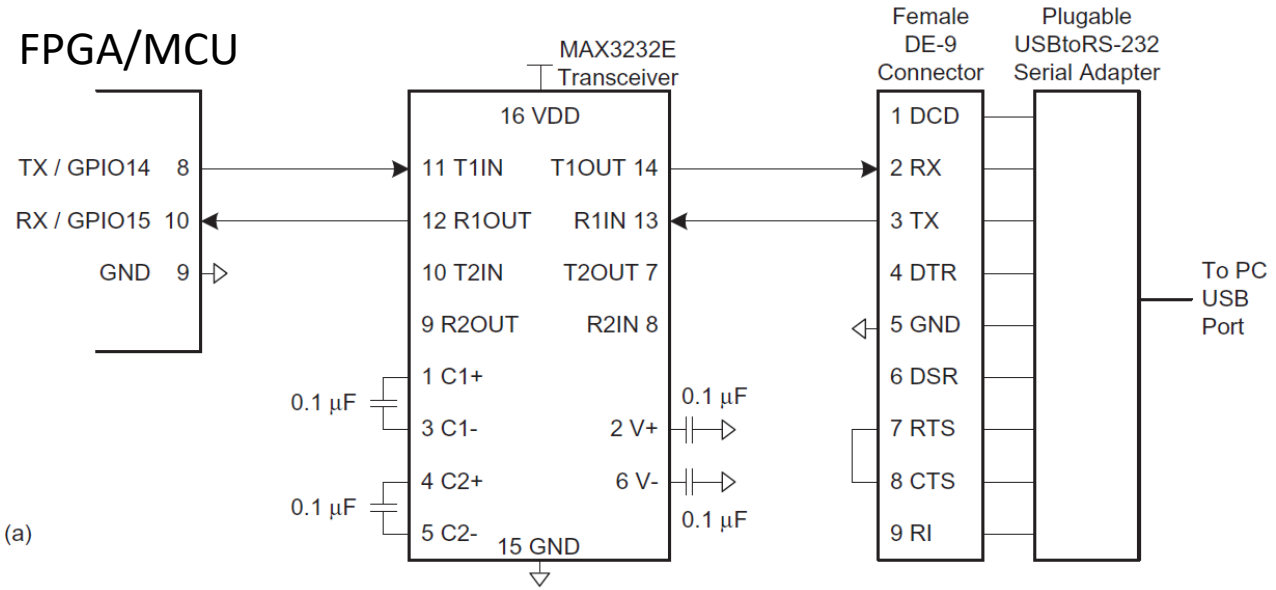
Data	
TXD (pin 3)	Serial Data Output
RXD (pin 2)	Serial Data Input
Handshake	
RTS (pin 7)	Request to Send
CTS (pin 8)	Clear to Send
DSR (pin 6)	Data Set Ready
DCD (pin 1)	Data Carrier Detect
DTR (pin 4)	Data Terminal Ready
Ground	
GND (pin 5)	Ground



DE-9 male cable in PC

UART: RS-232

- **Connecting to USB (RS-232)**
  - Sending text messages to PC←→ MCU
  - Typically use **MAX232**: match the same level of signal



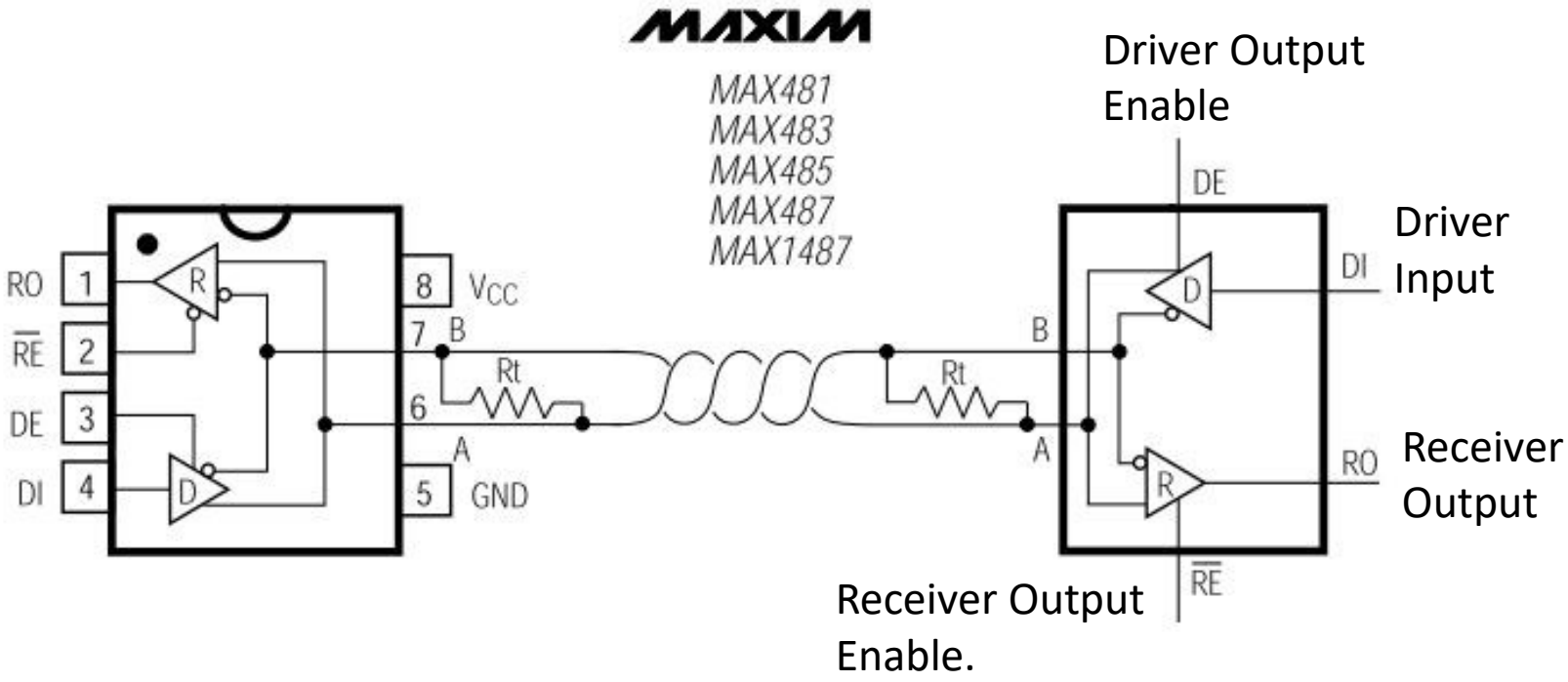


Commonly used ICs

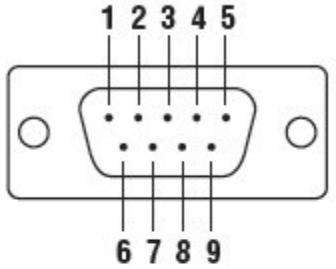
MAX14840E	40Mbps, +3.3V, RS-485 Half-Duplex Transceivers
MAX232	+5V-Powered, Multichannel RS-232 Drivers/Receivers
MAX3162	+3.0V to +5.5V, 1µA, RS-232/RS-485/422 Multiprotocol Transceivers
MAX3232	3.0V to 5.5V, Low-Power, Up to 1Mbps, True RS-232 Transceivers
MAX485	Low-Power, Slew-Rate-Limited RS-485/RS-422 Transceivers

UART: RS-485

- **UART: RS-485**
  - Use Differential voltage, more robust to noise RS-232
  - Can allow for greater distances
  - 100~120 ohm for load resistors (Rt)
  - Half-Duplex(two wires)
    - Receive or Send one at a time



DB9 male RS-422/485 port



PIN	RS-422/RS-485-4w	RS-485-2w
1	TxD+(B)	–
2	TxD-(A)	–
3	RxD+(B)	Data+(B)
4	RxD-(A)	Data-(A)
5	GND	GND
6	–	–
7	–	–
8	–	–

UART: RS-485

- **UART: RS-485**
  - Can allow up to 10 nodes
  - 100~120 ohm for load resistors (Rt)
  - RS-485 compatible with RS-422. But opposite is not true (Different voltage range)

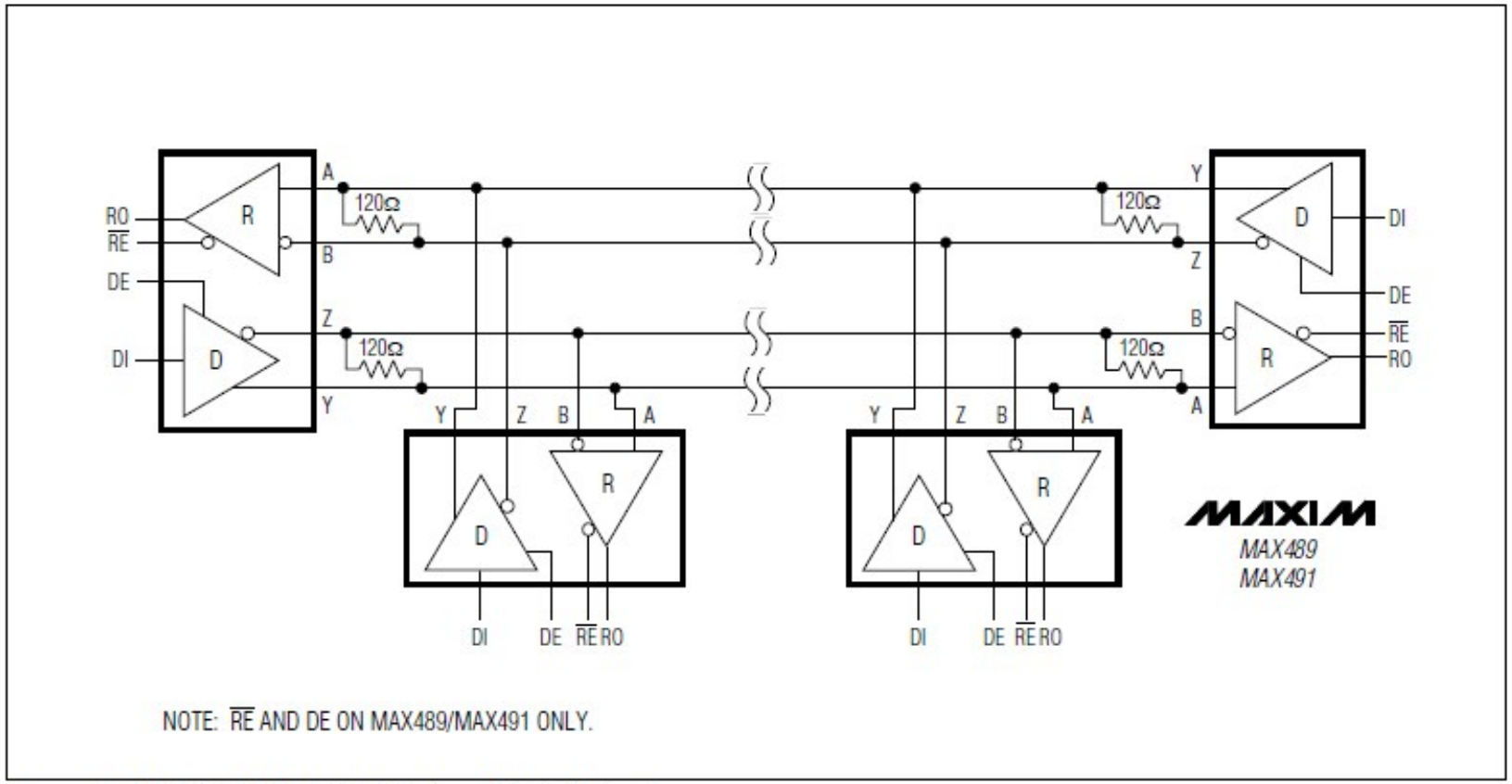
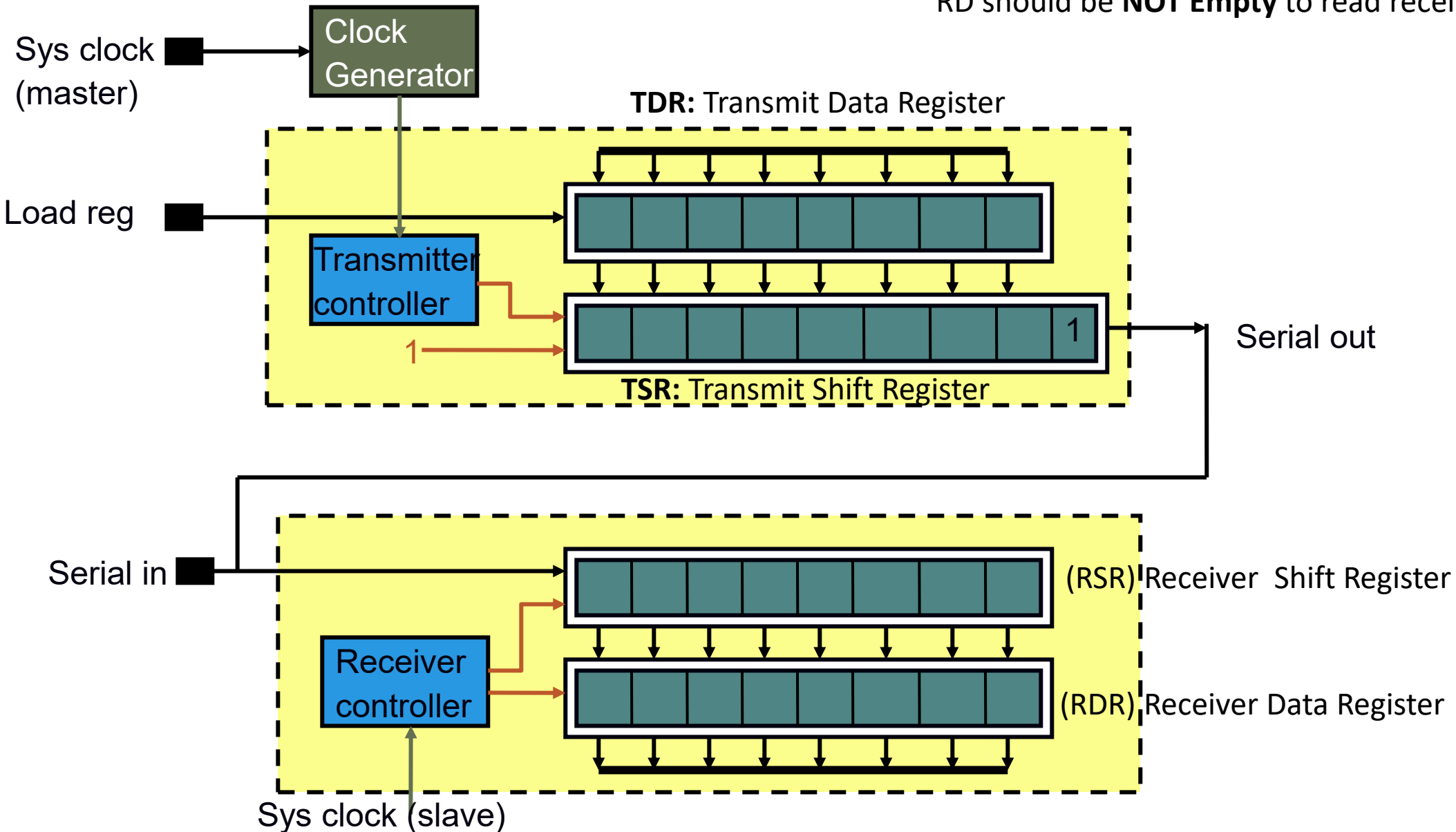


Figure 22. MAX488–MAX491 Full-Duplex RS-485 Network

UART Sending and Receiving Data

- **How to implement UART in Digital System**

TD should be **Empty** to send data : TXE=1
RD should be **NOT Empty** to read received data: RXNE=1

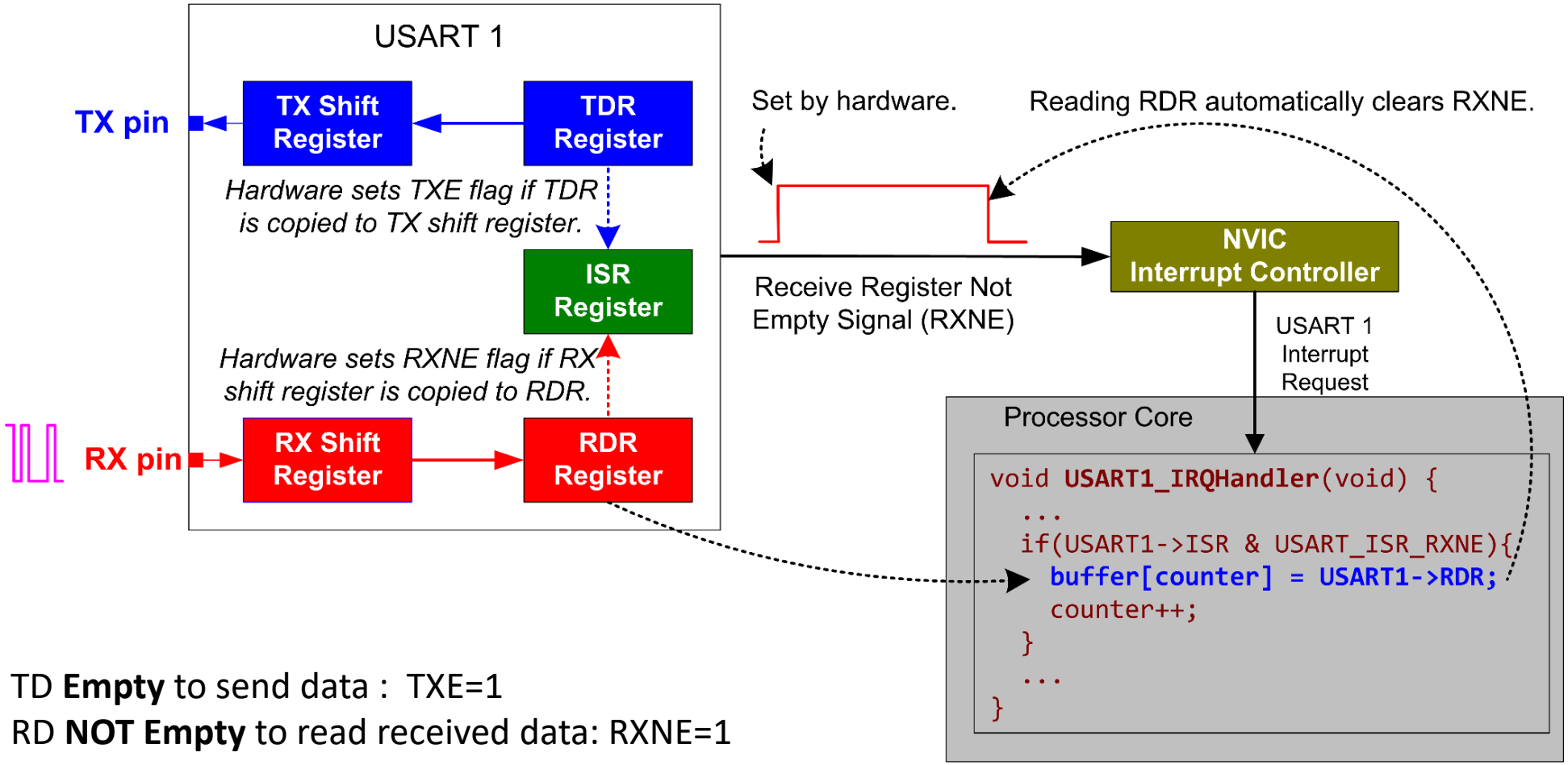


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UART Interrupt

UART Interrupt

- Polling vs Interrupt. Which is more burden to CPU?
- Usually setup as RX: Interrupt, TX: Polling



TD **Empty** to send data : TXE=1

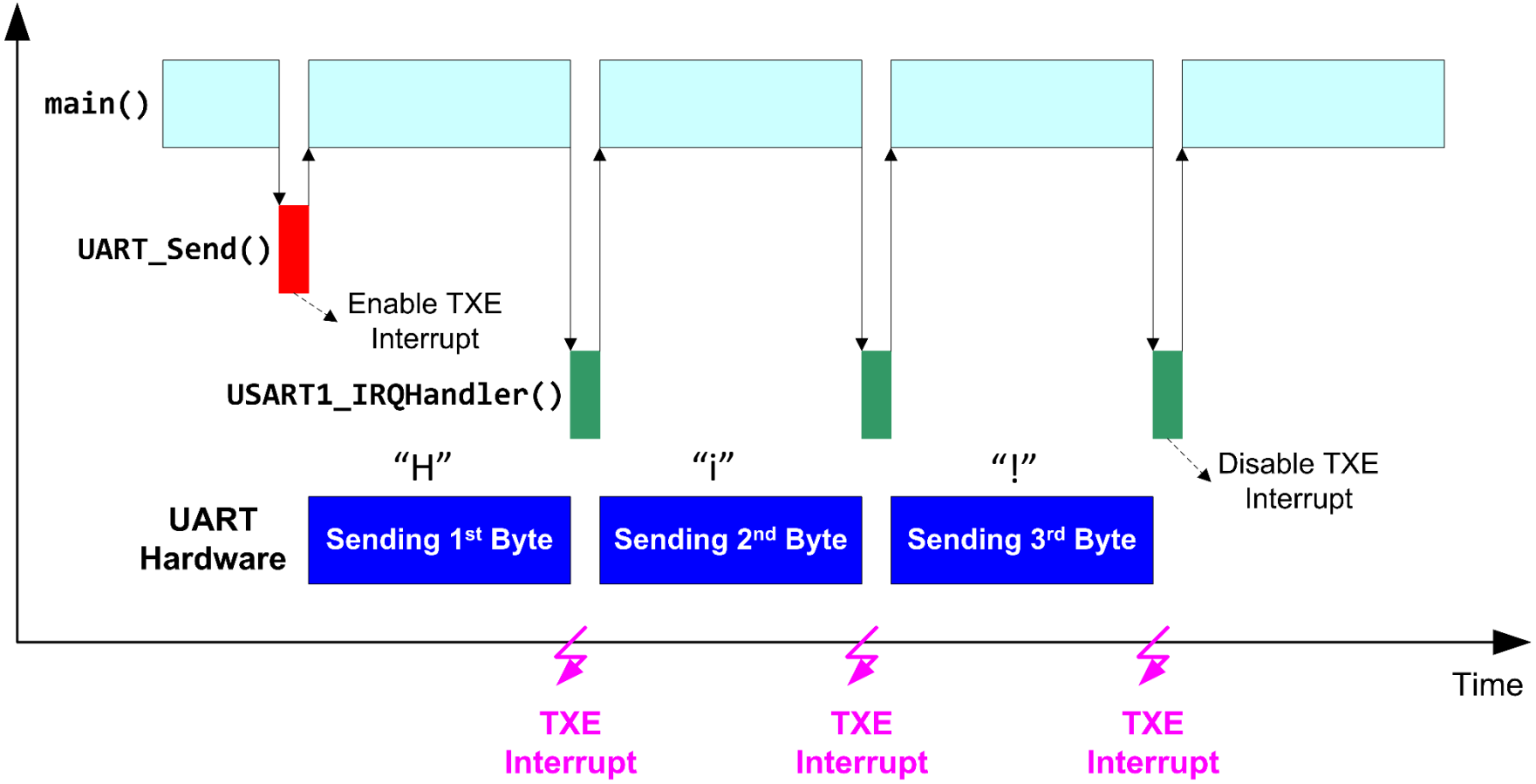
RD **NOT Empty** to read received data: RXNE=1

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UART Interrupt

UART Interrupt

- Example of sending 3 characters via interrupt : "Hi!"



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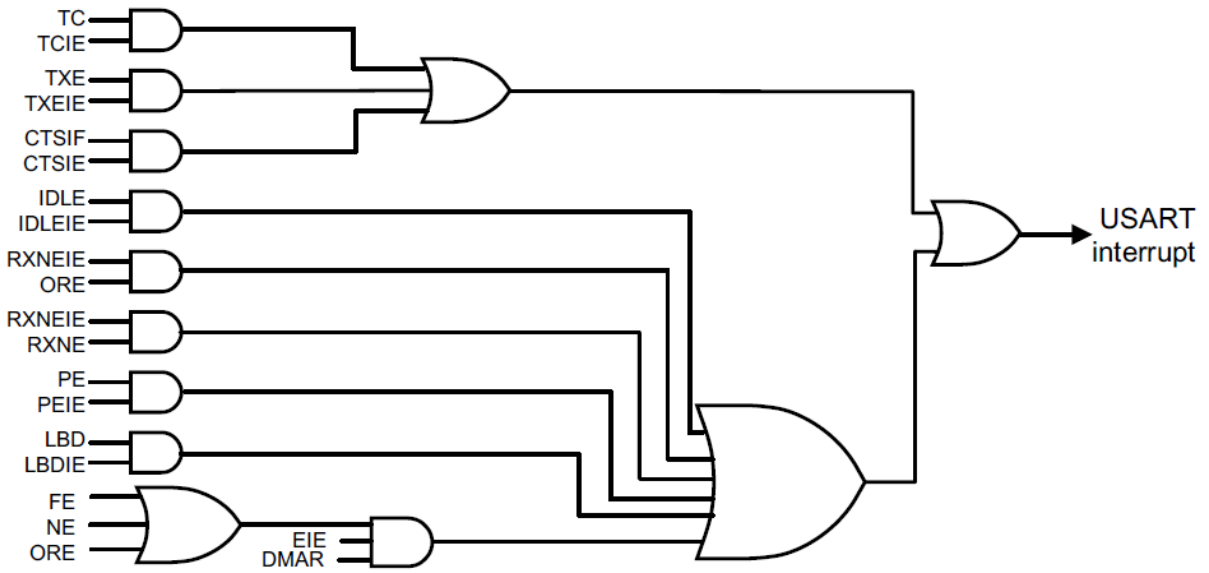
UART Interrupt

- **UART Interrupt**
  - For STM32f4x

Table 86. USART interrupt requests

Interrupt event	Event flag	Enable control bit
Transmit Data Register Empty	TXE	TXEIE
CTS flag	CTS	CTSIE
Transmission Complete	TC	TCIE
Received Data Ready to be Read	RXNE	RXNEIE
Overrun Error Detected	ORE	
Idle Line Detected	IDLE	IDLEIE
Parity Error	PE	PEIE
Break Flag	LBD	LBDIE
Noise Flag, Overrun error and Framing Error in multibuffer communication	NF or ORE or FE	EIE

Figure 191. USART interrupt mapping diagram



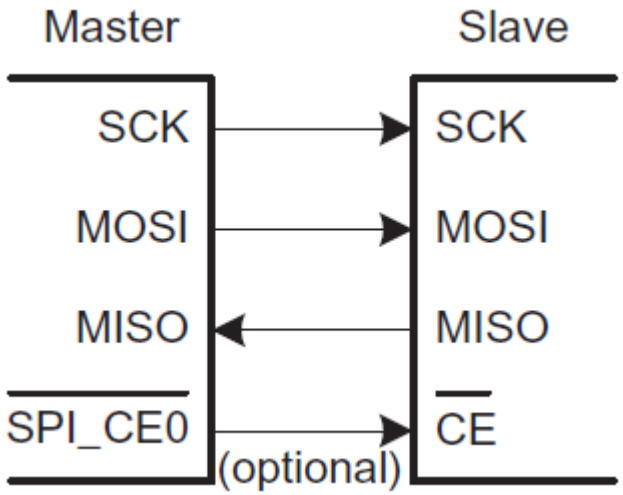
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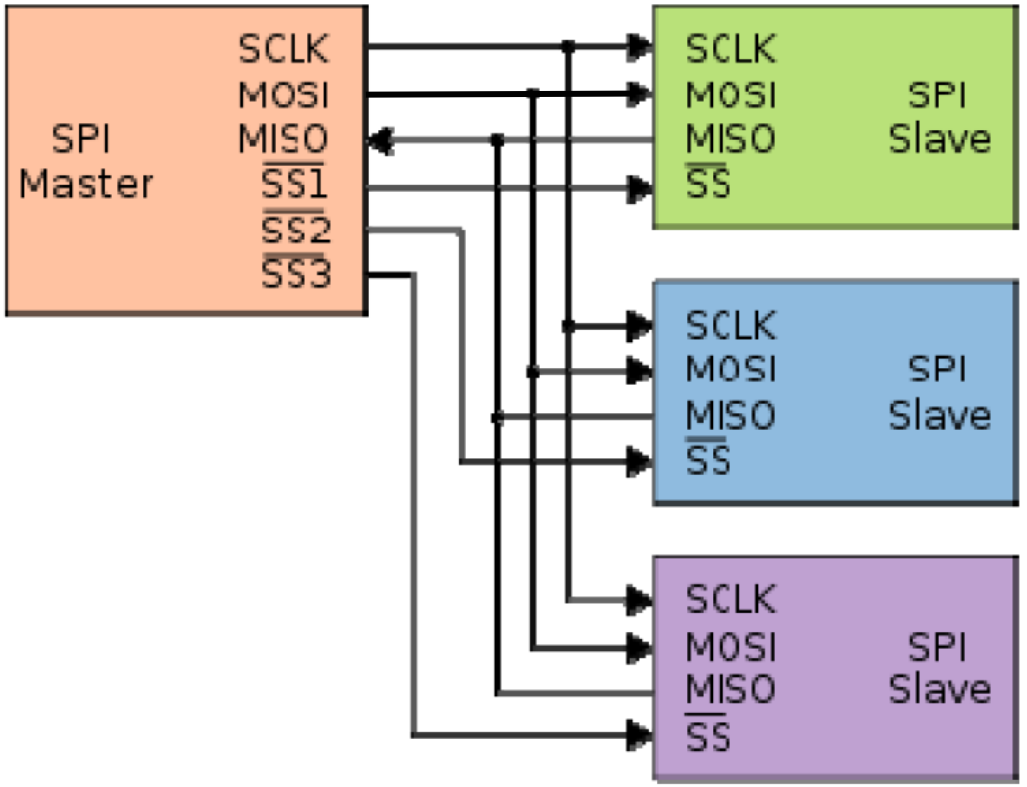
Synchronous: SPI

SPI (Serial Peripheral Interface)

- **SPI**
  - A simple **synchronous** serial protocol
  - Communication between Master and Slave
  - Faster rate than UART, can have multiple Slaves
  - Ports:
    - Serial Clock (SCK),
    - Master Out Slave In (MOSI, SDO)
    - Master In Slave Out (MISO, SDI).

SPI0CLK	SPI Frequency (kHz)
2	125000
4	62500
8	31250
16	15625
32	7812
64	3906

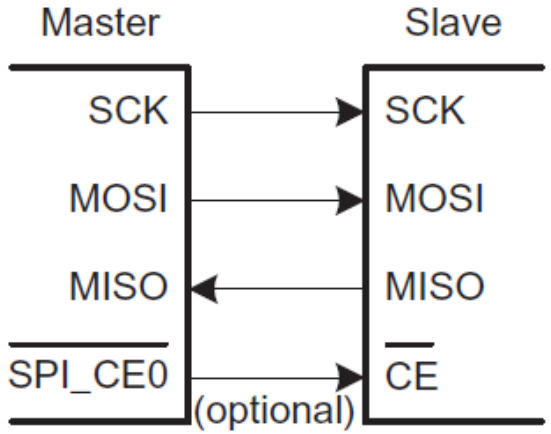


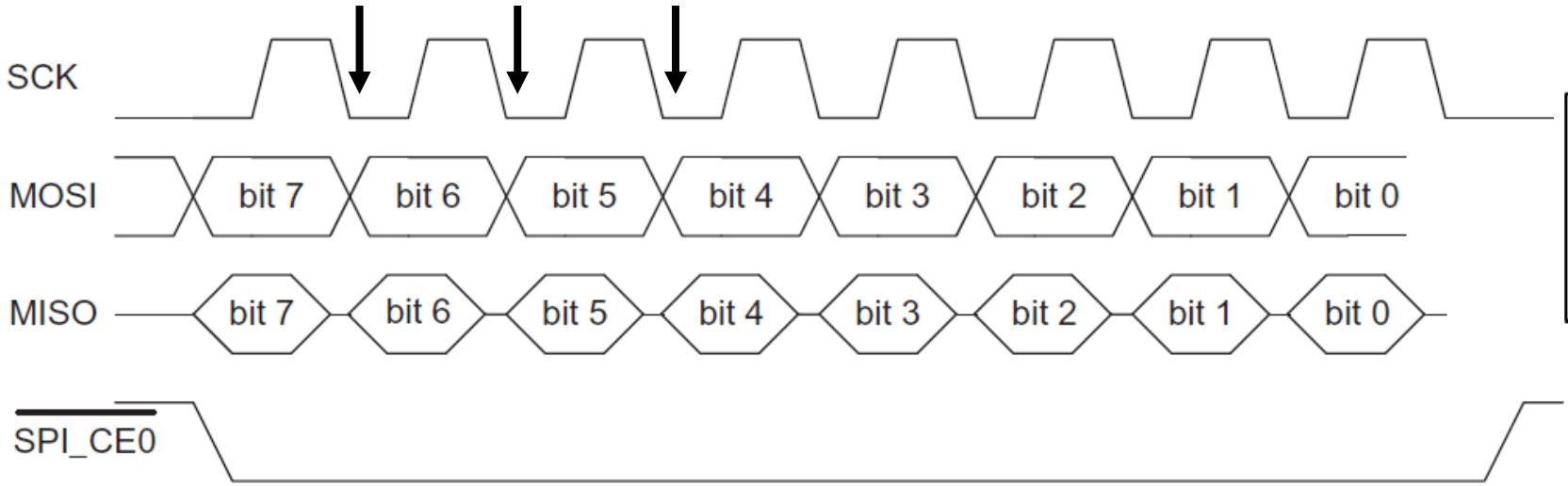


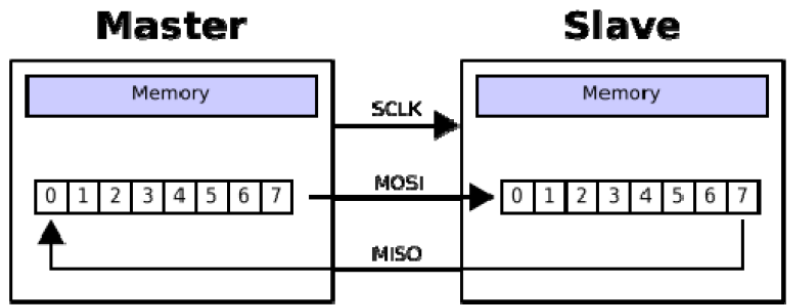
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SPI (Serial Peripheral Interface)

- **SPI**
  - Master unit generates the CLK
  - Initiates communication by sending a series of clock pulses on SCK
  - Bits change on the falling edge of SCK
  - Sending data from Master → Slave (MOSI)
    - MSB sent first
    - Slave may simultaneously respond by putting data on MISO
  - Chip Enable(active low CE): to alert the receiver that data is sent

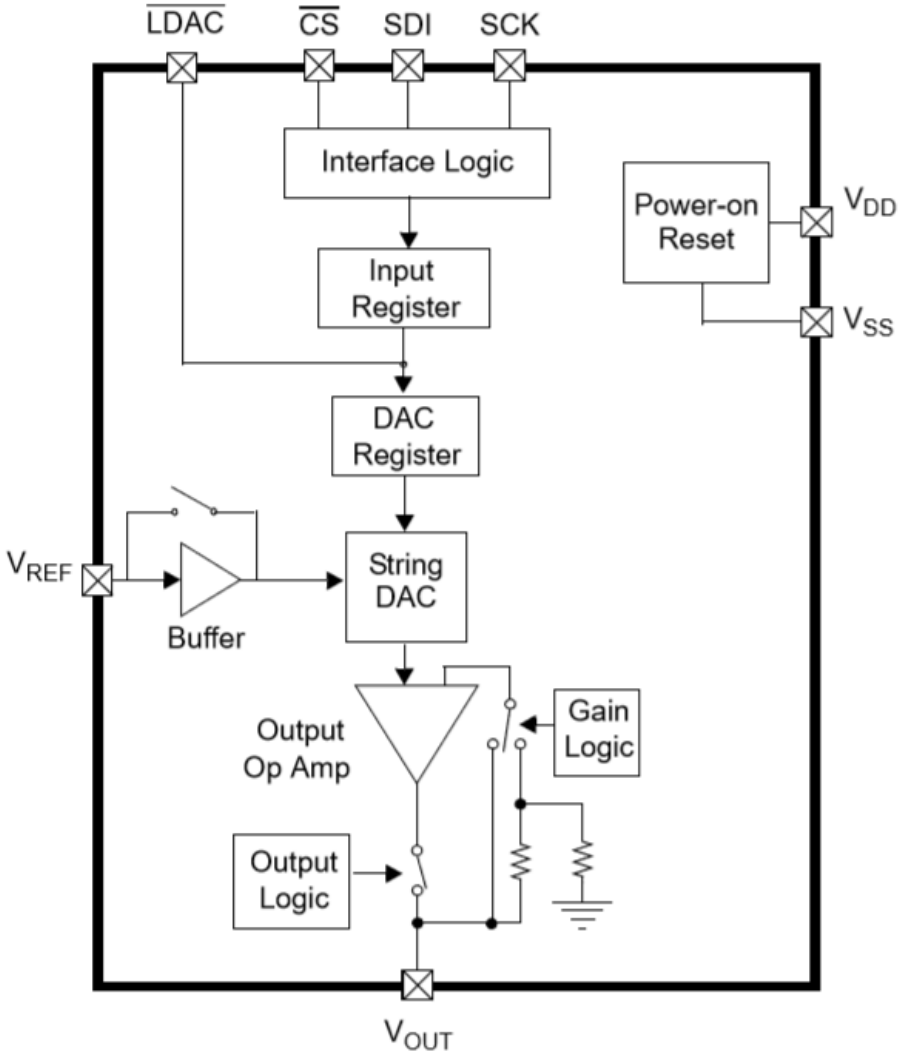






SPI (Serial Peripheral Interface)

SPI example: DAC, Microchip MCP4911





MCP4911 ☆

Status: In Production

View Datasheet

Features:

- 10-bit Resolution

MICROCHIP **MCP4901/4911/4921**

8/10/12-Bit Voltage Output Digital-to-Analog Converter with SPI Interface

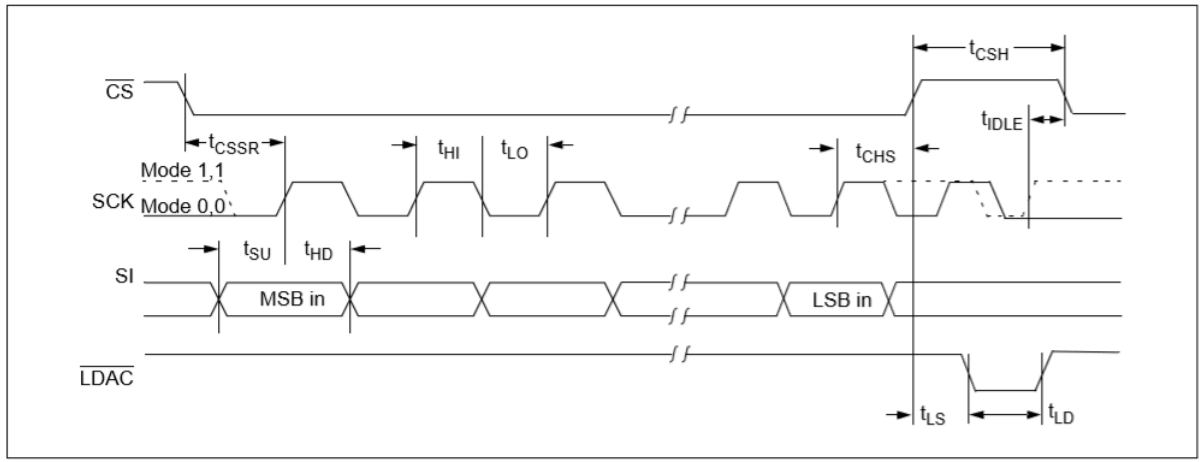


FIGURE 1-1: *SPI Input Timing Data.*

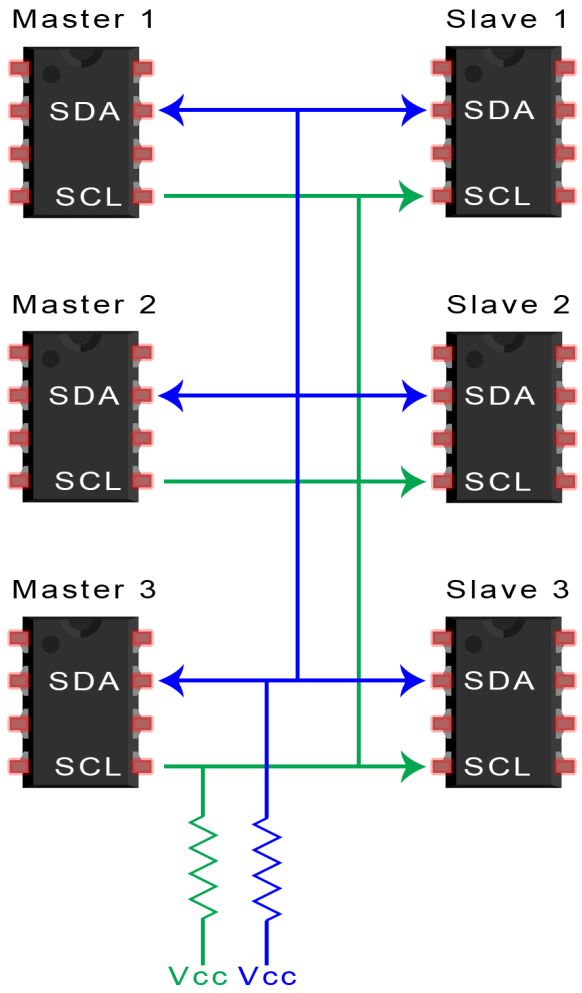
https://www.microchip.com/wwwproducts/en/MCP4911

I2C (Inter-Integrated Circuit)

I2C (Inter-Integrated Circuit)

- A serial communication protocol
- Combined features of SPI and UART
  - Allow Multi-Master, Multi-Slave connection (SPI: single master)
  - Synchronous like SPI
  - Use 2 wires like UART
- Usually 100kbps, but can go as high as 5Mbps

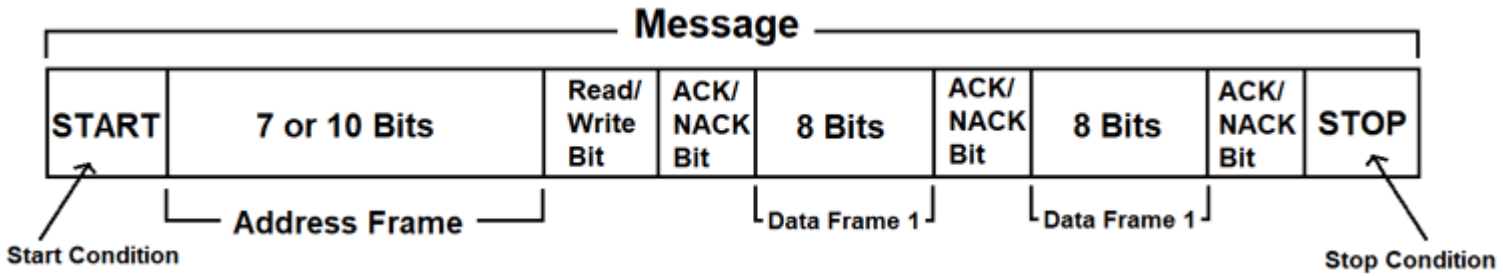
- SDA: Serial Data
  - Send/Receive data line
- SCL: Serial Clock
  - CLK signal



I2C (Inter-Integrated Circuit)

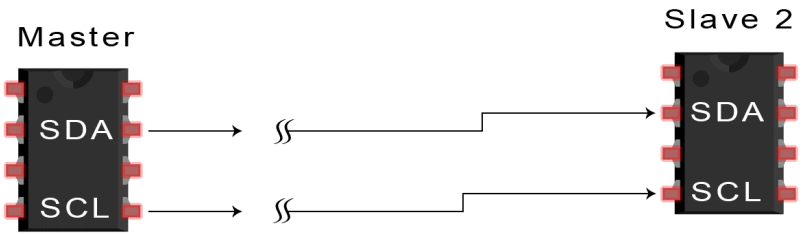
- **I2C (Inter-Integrated Circuit)**
  - Transfer data in messages: frames of data
  - Contains address of the slave, start, stop, read/write, ACK/NACK
  - Each message can contain several data frames(8bits)

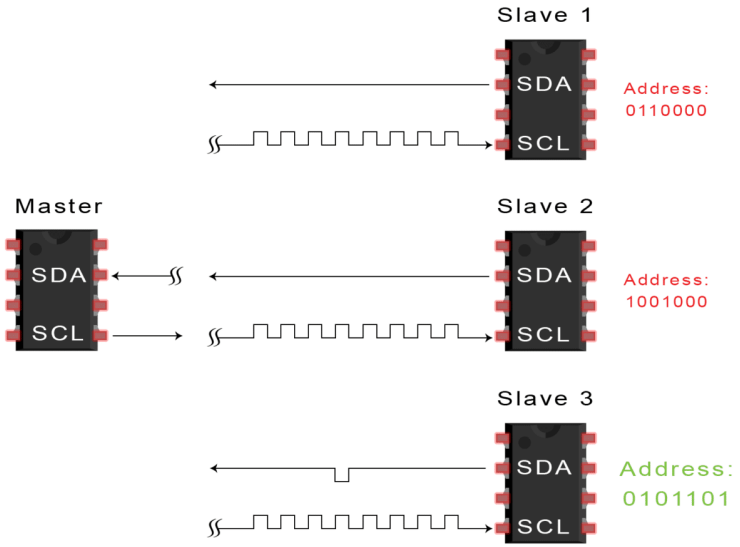
  - Start: 'H' → 'L' before SCL falling edge
  - Stop: 'L' → 'H' after SCL rising edge
  - Address: ID of slave
  - Read/Write: Master Sending('H'), Receiving ('L')
  - ACK/NACK: For each data frame received successfully, ACK is received

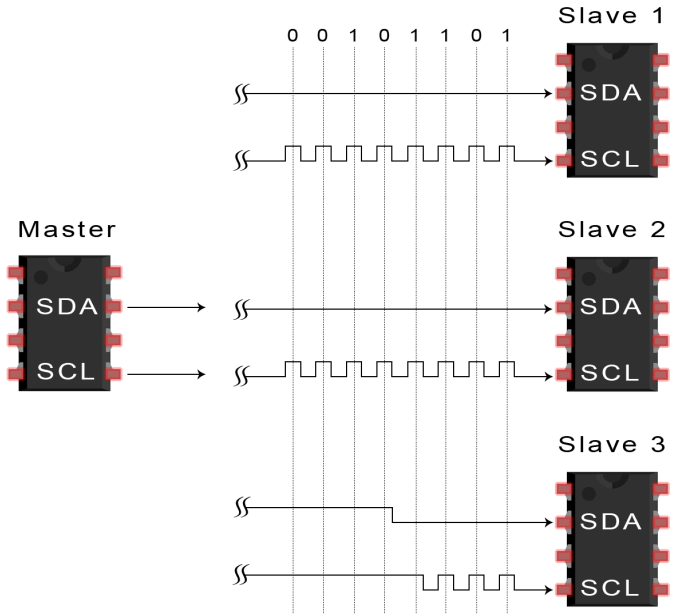


I2C (Inter-Integrated Circuit)

- **I2C (Inter-Integrated Circuit)**
  1) Start: 'H' → 'L' *before* SCL falling edge
  2) All slaves receive data (start+address +R/W bits)
  3) Matched Slave sends ACK bits (SDA 'H'->'L')
  4) Master sends or receives data frames
  5) Every data frame, slave sends ACK bits
  6) To Stop, 'L' to 'H' after switching SCL rising edge

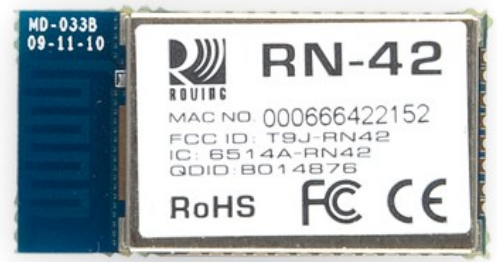


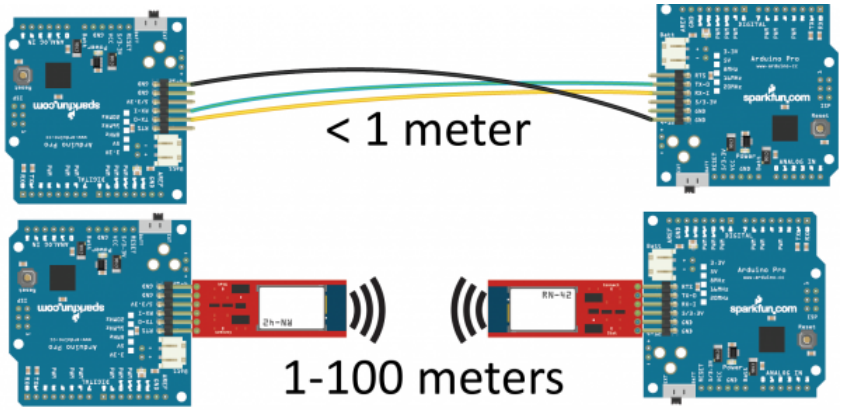




Bluetooth

- **Bluetooth**
  - A wireless serial communication protocol
    - Invented by Ericsson 1994, Maintained by SIG: Special Interest Group
    - Short range connectivity, license free spec.
    - Very low powered. A master can be connected to several slaves
  - Band 2.4GHz
    - Industrial, Scientific and Medical Frequency band
  - Unique 48-bit address (BD_ADDR)





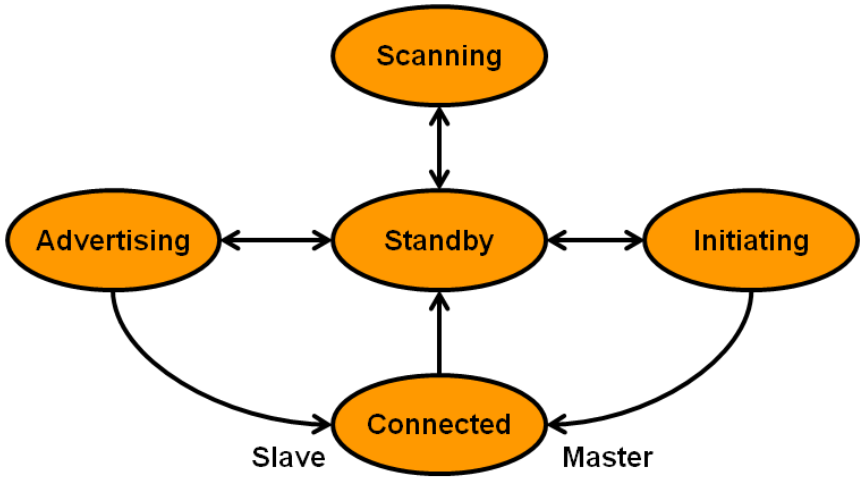
BLUETOOTH VERSION COMPARISON

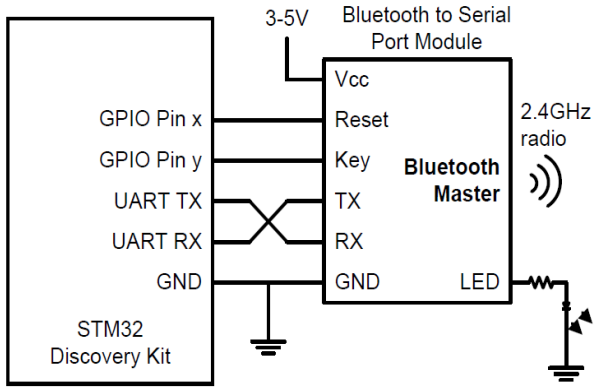
Key Features, Speed & Range (1.x, 4.x, 5.x, 6.x)

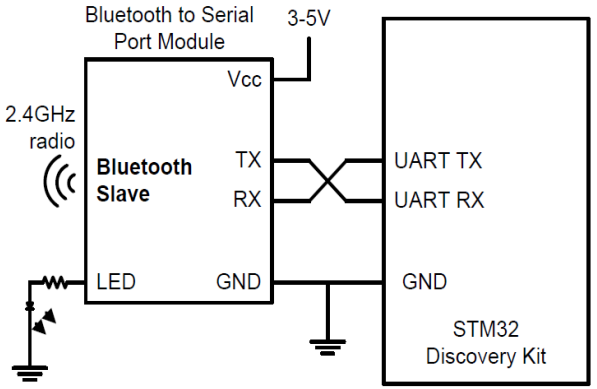
VERSION	KEY FEATURES & IMPROVEMENTS	MAX SPEED (Theoretical)	MAX RANGE (Theoretical)
BLUETOOTH 1.x	Early standard, basic wireless data transfer, first audio/headset connectivity.	Up to 1 Mbps	~10 meters
BLUETOOTH 4.x (BLE)	Introduced Bluetooth Low Energy (BLE), optimized for wearables and IoT, Beacon support.	Up to 1 Mbps	~50 meters
BLUETOOTH 5.x	Major upgrade vs 4.x (2x Speed, 4x Range, 8x Data capacity), Direction Finding, and LE Audio.	Up to 2 Mbps↑	~200 meters
BLUETOOTH 6.x (Latest)	Channel Sounding (cm-level precise distance tracking), ultra-low latency, and enhanced security.	Up to 2 Mbps	~200 meters

Bluetooth

- Communication
  - Exchange data in one of two states
    - Advertising and Connected modes
  - Standby -> Discovery -> Connecting -> Connected







Others Radio Frequency (RF)

nRF24L01 (uses SPI)

